

WILLIAM NGUYEN

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EDUCATION

San Jose State University

B.S. Software Engineering, GPA: 3.26

San Jose, CA

Expected May 2027

- Relevant Coursework: Embedded Systems, Computer Architecture, Operating Systems, Control Systems

PROJECTS

STM32G474 Custom UART Bootloader | Embedded C, Python | [\[GitHub\]](#)

- Built a Cortex-M4 UART bootloader with 64 KB boot / 448 KB application flash partitioning for safe over-the-wire firmware updates.
- Designed a 512-byte image header with boot handoff via VTOR remap, MSP reload, reset-vector validation, and branch to Reset_Handler.
- Added CRC32 verification, MSP range checks, and Thumb-bit validation at boot with bounded erase/write to prevent bootloader corruption.
- Wrote a Python serial host tool for end-to-end flash programming and update testing over UART.

FreeRTOS Sensor Hub Platform | C, FreeRTOS, STM32G474 | [\[GitHub\]](#)

- Built an RTOS-based STM32 sensor hub with telemetry, CLI, and peripheral bring-up for repeatable board validation.
- Structured firmware into application, services, modules, and utilities, separating handwritten logic from generated HAL code.
- Integrated FreeRTOS tasks for CLI, telemetry, and GPIO control with retargeted printf over LPUART1 at 115200 8N1; brought up I2C, BOOT output, 500 ms heartbeat, and ~10 Hz telemetry.

YOKO Embedded Robotic Hand | C/C++, ESP32 | [\[GitHub\]](#)

- Developed ESP32 firmware to drive 5 DC motors with 1 kHz PWM for a five-finger robotic hand, enabling stable per-finger actuation.
- Built modular motor-control, calibration, and safety components with homing state machines and travel limits stored in non-volatile memory.
- Integrated FSR fingertip sensing with over-current/stall cutoff protection, achieving 18/20 successful grasp trials.

EXPERIENCE

Embedded Systems Contributor

Robotics Club – San Jose State University

San Jose, CA

Aug 2025 – Present

- Debugged I2C and SPI peripheral bring-up across sensor and motor-driver boards using logic analyzer and serial decoding, reducing team integration time.
- Probed signal integrity, verified power rail sequencing, and isolated firmware faults during board-level hardware-software integration.
- Created wiring diagrams and structured test notes in Git, improving build repeatability and reducing rework across hardware revisions.

Independent Embedded Developer

Self-Directed – Personal & Open-Source

San Jose, CA

Jan 2025 – Present

- Architected and shipped 3 end-to-end embedded projects (STM32, ESP32) with version-controlled source, build systems, and documented READMEs on GitHub.
- Wrote bare-metal peripheral drivers (UART, I2C, GPIO, PWM) and a custom linker script for flash partitioning, going beyond HAL to register-level interaction.
- Built reusable firmware modules (CLI shell, telemetry logger, FreeRTOS task scaffolding) shared across personal projects and the SJSU Robotics Club codebase.

TECHNICAL SKILLS

Languages: C, C++, Python

Firmware/Embedded: STM32, ESP32, FreeRTOS, Bootloaders, Linker Scripts, Bare-Metal Drivers, State Machines, Flash Memory

Protocols & Interfaces: UART/Serial, I2C, SPI, GPIO, PWM, CRC32, JTAG/SWD

Tools & Debugging: Git, Linux, GDB, Logic Analyzer, STM32CubeIDE, PlatformIO, ESP-IDF, HAL/CMSIS, VS Code